Aryan Agarwal

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Education

University of Illinois Urbana-Champaign **Champaign, IL**

Major: Computer Engineering (GPA: 3.89) *Expected Graduation: May 2028*

Relevant Coursework: Autonomous Vehicles, Operating Systems, Robotic Systems, Real-Time Systems, Signal Processing, Data Structures and Algorithms, Embedded DSP, Parallel Programming

Experience

Molex **May 2026 – Aug 2026**

Firmware Engineering Intern *Fremont, CA*

- Automated Molex's firmware codebase migration from Keil µVision to SEGGER Embedded Studio.
- Developed **embedded C firmware** for a 1PPS synchronization system using hardware-timer capture, interrupt-driven timestamping, and EEPROM-stored calibration data to achieve **sub-nanosecond timing accuracy**.
- Enabled a **shared control loop** to support two vendor analog front ends through **runtime chip-ID detection**, **dynamic driver selection**, and **device-specific register configuration**.
- Redesigned Molex's firmware codebase to support multiple MCUs by separating shared logic from platform-specific logic.
- Added platform-specific **FreeRTOS** support and **I2C** and **SPI** peripheral code for multi-MCU operation.

Cisco **May 2025 – Aug 2025**

ASIC Engineering Intern *San Jose, CA*

- Measured **SDF-annotated** delays from inputs to flip-flop D-pins and clock pins across **thousands of signals** using a **Python-scripted SystemVerilog testbench**.
- Computed per-signal skew offsets and applied them through an RTL wrapper, achieving **100% gate-level vs RTL alignment**.
- Developed an **FSDB** waveform analysis tool to compare expected vs. observed signal transitions, reducing manual debug time by **35%** and accelerating root-cause analysis of gate-level timing failures.
- Designed an **HTML** report interface with searchable signal-timestamp tables, reducing waveform inspection time by **25%**.

Spatial Audio Research, UIUC **Jan 2026 – Present**

Undergraduate Researcher – Advised by Prof. Yu-Lin Wei *Champaign, IL*

- Kept sound sources world-locked during live head motion by combining orientation data from a **9-axis IMU** with a real-time **Python binaural renderer** using CIPIC **HRIRs** and **SciPy FFT-based convolution**.
- Reduced audible artifacts during head motion by replacing **nearest-neighbor HRIR selection** with **barycentric interpolation** across neighboring CIPIC measurements, smoothing filter updates as the listener's head orientation changed.
- Improved perceived motion-to-audio latency by using **STFT** analysis of recent IMU measurements to extrapolate head motion **100 ms** ahead and proactively update the corresponding **HRIR filters**.

Projects

Vision-Guided UR3 Drawing Robot

- Developed an image-to-motion pipeline that turns input images into paper drawings by extracting **OpenCV** contours, mapping pixels to calibrated workspace coordinates, and generating **UR3** trajectories with custom **FK/IK**.
- Generated drawing waypoints by applying polygonal simplification to contours extracted using median filtering and edge detection.
- Used **ROS** to execute MoveJ and MoveL commands across generated waypoints, accurately reproducing input images.

VisionVest - Won 2nd Place at World's Largest Hardware Hackathon

- Developed **ESP32 firmware** for a BLE haptic vest, translating AI perception commands into real-time motor actuation.
- Implemented a **9-field BLE JSON protocol** with packet timeouts, priority arbitration, and mode-specific control.
- Integrated **ultrasonic proximity sensing** to trigger local obstacle safety overrides independent of iPhone perception.
- Validated **4 navigation modes**: awareness, object search, GPS guidance, and manual haptic testing.

RISC-V Operating System Kernel

- Built a **RISC-V OS kernel** in **C** supporting user-mode execution, threading, traps, syscalls, and process management.
- Implemented **VirtIO-backed storage** with NGFS filesystem support, block caching, persistent I/O, and directory operations.
- Developed **Sv39 virtual memory** with 3-level page-tables, process address-space cloning, and lazy page allocation.
- Integrated ELF loading, scheduling, interrupt/exception handling, **UART I/O**, and shell support for disk program execution.

CUDA GPT-2 Inference Optimizer

- Optimized kernels for inference across attention, layernorm, softmax, GELU, residuals, and matmul, achieving **9.7x speedup**.
- Profiled workloads with nsys and ncu, analyzing kernel timing, memory traffic, cache behavior, and occupancy to find bottlenecks.

Skills

Programming Languages: C, C++, Python, CUDA, SystemVerilog, PyTorch, RISC-V Assembly, Java

Embedded and Firmware: STM32, FreeRTOS, ESP32, Raspberry Pi, I2C, SPI, CAN, UART, BLE, JTAG, SWD

Robotics: ROS, OpenCV, Kinematics, Sensor Fusion, IMUs, Motor Control, Epipolar Geometry, UR3

Tools: Linux, Git, GDB, Synopsys VCS, Verdi, DVE, QEMU, Keil µVision, SEGGER Embedded Studio, Vivado, KiCad, NumPy, SciPy, Jupyter Notebook, Catch2, LTspice, Oscilloscope, Logic Analyzer